

Math 141, S06

Exam 2

Section:

Name: *key*

SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	12	
2	12	
3	20	
4	24	
5	14	
6	18	
Total	100	

Good Luck !

Problem 1. (12 points)

(a) Let $f(x) = \frac{1}{x+1}$. Find $f'(2)$ using the definition of the derivative $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

$$\begin{aligned} \text{Solution: } f'(2) &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{2+h+1} - \frac{1}{2+1}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{3 - (3+h)}{3(3+h)}}{h} = \lim_{h \rightarrow 0} \frac{\frac{-h}{3(3+h)}}{h} \\ &= \lim_{h \rightarrow 0} -\frac{1}{3(3+h)} \\ &= -\frac{1}{9} \end{aligned}$$

(b) Determine whether the function $f(x) = x^7 + 10x$ is one-to-one by examining $f'(x)$. Explain.

$$\begin{aligned} \text{Solution: } f'(x) &= 7x^6 + 10 \geq 10 > 0 \text{ for any } x \\ \text{So } f(x) &\text{ is one-to-one on } (-\infty, +\infty) \end{aligned}$$

Problem 2. (12 points) A Folium of Descartes is defined implicitly by the equation $x^3 + y^3 = 3xy$.

(a) Use the implicit differentiation to find $\frac{dy}{dx}$.

Solution:

$$\begin{aligned}\frac{d}{dx}[x^3 + y^3] &= \frac{d}{dx}[3xy] \\ 3x^2 + 3y^2 \frac{dy}{dx} &= 3y + 3x \frac{dy}{dx} \\ (3y^2 - 3x) \frac{dy}{dx} &= 3y - 3x^2 \\ \frac{dy}{dx} &= \frac{3y - 3x^2}{3y^2 - 3x} \\ &= \frac{y - x^2}{y^2 - x}\end{aligned}$$

(b) Find an equation for the tangent line to the Folium of Descartes at the point $\left(\frac{3}{2}, \frac{3}{2}\right)$ using the result from (a).

Solution: Since

$$\left. \frac{dy}{dx} \right|_{\left(\frac{3}{2}, \frac{3}{2}\right)} = \frac{3 \cdot \left(\frac{3}{2}\right) - 3 \cdot \left(\frac{3}{2}\right)^2}{3 \cdot \left(\frac{3}{2}\right)^2 - 3 \cdot \left(\frac{3}{2}\right)} = -1$$

We have the slope of the tangent line equals -1 .

Then, the equation is

$$y - \frac{3}{2} = (-1) \cdot \left(x - \frac{3}{2}\right)$$

$$y = -x + \frac{3}{2} + \frac{3}{2}$$

$$y = -x + 3$$

Problem 3. (20 points) Find each requested derivative. Be sure to simplify your results.

(a) Find $f'(x)$ when $f(x) = \frac{x}{x^2+9}$

$$\begin{aligned} f'(x) &= \frac{1 \cdot (x^2+9) - x \cdot (2x)}{(x^2+9)^2} \\ &= \frac{x^2+9-2x^2}{(x^2+9)^2} = \frac{-x^2+9}{(x^2+9)^2} \end{aligned}$$

(b) Find $\frac{dy}{dx}$ when $y = x \tan x^2$

$$\begin{aligned} \frac{dy}{dx} &= x \cdot (\sec^2 x^2) \cdot (2x) + \tan x^2 \\ &= 2x^2 \sec^2 x^2 + \tan x^2 \end{aligned}$$

(c) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $y = \sin x \cos x$

$$\frac{dy}{dx} = -\sin^2 x + \cos^2 x = \cos 2x$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= -2 \sin x \cos x + 2 \cos x (-\sin x) \\ &= -4 \sin x \cos x \end{aligned}$$

Problem 4. (24 points) Find the derivative of each of the following functions.

(a) $f(x) = x^2 \ln x$

$$\begin{aligned} f'(x) &= x^2 \cdot \left(\frac{1}{x}\right) + (2x) \ln x \\ &= x + 2x \ln x \end{aligned}$$

(b) $f(x) = e^{x \sin x}$

$$\begin{aligned} f'(x) &= e^{x \sin x} \cdot \frac{d}{dx} [x \sin x] \\ &= e^{x \sin x} \cdot (x \cos x + \sin x) \end{aligned}$$

(c) $f(x) = 3^{-\sqrt{x}}$

$$\begin{aligned} f'(x) &= 3^{-\sqrt{x}} (\ln 3) \cdot \frac{d}{dx} [-\sqrt{x}] \\ &= -\frac{3^{-\sqrt{x}} \ln 3}{2\sqrt{x}} \end{aligned}$$

(d) $f(x) = \tan^{-1}(x^2)$

$$\begin{aligned} f'(x) &= \frac{1}{1+(x^2)^2} \cdot 2x \\ &= \frac{2x}{1+x^4} \end{aligned}$$

Problem 5. (14 points) Using the logarithmic differentiation to find the $\frac{dy}{dx}$ if

$$(a) y = \frac{(x^2 + 2)^{3/2} \sqrt{x^2 + 1}}{x^3 + 4x}$$

Solution: $\ln y = \ln \left[\frac{(x^2 + 2)^{3/2} \sqrt{x^2 + 1}}{x^3 + 4x} \right]$

$$= \frac{3}{2} \ln(x^2 + 2) + \frac{1}{2} \ln(x^2 + 1) - \ln(x^3 + 4x)$$

$$\frac{d}{dx} [\ln y] = \frac{3}{2} \frac{1}{x^2 + 2} \cdot (2x) + \frac{1}{2} \frac{1}{x^2 + 1} \cdot (2x) - \frac{1}{x^3 + 4x} \cdot (3x^2 + 4)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{3x}{x^2 + 2} + \frac{x}{x^2 + 1} - \frac{3x^2 + 4}{x^3 + 4x}$$

$$\frac{dy}{dx} = \frac{(x^2 + 2)^{3/2} \sqrt{x^2 + 1}}{x^3 + 4x} \left(\frac{3x}{x^2 + 2} + \frac{x}{x^2 + 1} - \frac{3x^2 + 4}{x^3 + 4x} \right)$$

$$(b) y = (x^2 + 1)^{\cos x}$$

Solution: $\ln y = \cos x \ln(x^2 + 1)$

$$\frac{d}{dx} [\ln y] = \frac{\cos x}{x^2 + 1} \cdot 2x + (-\sin x) \ln(x^2 + 1)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{2x \cos x}{x^2 + 1} - \sin x \ln(x^2 + 1)$$

$$\frac{dy}{dx} = (x^2 + 1)^{\cos x} \left[\frac{2x \cos x}{x^2 + 1} - \sin x \ln(x^2 + 1) \right]$$

Problem 6. (18 points) Evaluate the following limits. Identify each time l'Hopital's Rule is applied, including the type of indetermined form.

(a) $\lim_{x \rightarrow 2} \frac{x^3 - 2x - 4}{x^2 - x - 2}$

$$\lim_{x \rightarrow 2} x^3 - 2x - 4 = 0$$

$$\lim_{x \rightarrow 2} x^2 - x - 2 = 0$$

$\frac{0}{0}$ type

$$\lim_{x \rightarrow 2} \frac{x^3 - 2x - 4}{x^2 - x - 2} = \lim_{x \rightarrow 2} \frac{3x^2 - 2}{2x - 1} = \frac{10}{3}$$

(b) $\lim_{x \rightarrow 0^+} \frac{\cos x}{\ln x}$

$$\lim_{x \rightarrow 0^+} \cos x = 1$$

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

So l'Hopital's Rule isn't applied here.

$$\lim_{x \rightarrow 0^+} \frac{\cos x}{\ln x} = 0$$

(c) $\lim_{x \rightarrow +\infty} \frac{x^3}{e^{2x}}$

$$\lim_{x \rightarrow +\infty} x^3 = +\infty$$

$$\lim_{x \rightarrow +\infty} e^{2x} = +\infty$$

$\frac{\infty}{\infty}$ type

$$\lim_{x \rightarrow +\infty} \frac{x^3}{e^{2x}} = \lim_{x \rightarrow +\infty} \frac{3x^2 \rightarrow +\infty}{2e^{2x} \rightarrow +\infty}$$

$$= \lim_{x \rightarrow +\infty} \frac{6x \rightarrow +\infty}{4e^{2x} \rightarrow +\infty}$$

$$= \lim_{x \rightarrow +\infty} \frac{6}{8e^{2x} \rightarrow +\infty}$$

$$= 0$$